HIGH-LEVEL DESIGN

Body Control Module - Door and Access Control High-Level Design

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Project: PRJ-BODY
Classification: Synthetic training data - no real vehicle program

1 Introduction

This document describes the high-level software design of the door and access control functions of the Body Control Module (BCM). It is written for the AUTOSAR Classic Platform and covers the application software components, their ports and interfaces, the signals exchanged between them and the main functional flows.

The design is synthetic and was created only for training and evaluation of an AI document analysis assistant. It does not describe a real vehicle.

2 System Overview

The BCM controls the central locking of four doors. Lock and unlock requests originate from the key fob, from the speed-based automatic locking function and from the crash unlock function. All requests are arbitrated in one software component and executed by a separate actuator control component.

3 Architecture Overview

The software follows the AUTOSAR layered architecture. Application software components communicate exclusively through the Runtime Environment (RTE) using Sender-Receiver interfaces. Hardware access is provided by the Basic Software (BSW) with the CAN stack for network communication and the I/O hardware abstraction for the door lock actuators.

The operating system schedules runnables in periodic tasks of 10 ms, 20 ms, 50 ms and 100 ms.

4 Software Component Design

This section describes every application software component with its ports, dependencies and verification cases.

4.1 SWC_KeylessEntryMgr

Component: SWC_KeylessEntryMgr

Type: Application SWC

Safety Level: ASIL B

Description: Decodes authenticated key fob and passive-entry requests and publishes a single key command for the door locking subsystem. It filters repeated RF frames and debounces the request.

Design Rationale: A dedicated manager isolates radio-frequency handling from the lock arbitration logic so that the arbitration can be reused for other request sources.

Runnables: Run_KeylessEntryMgr_50ms (50 ms)

Provided Ports:

```
- P_KeyCmd | SR_KeyFobCommand | Sender-Receiver | period 50 ms
```

Required Ports:

```
- none
```

Depends On: SWC_ImmobilizerAuth

Error Handling: If the RF receiver reports a fault the key command is set to INVALID and the request is ignored.

Verification: TC_BCM_001, TC_BCM_002

4.2 SWC_CentralLockArb

Component: SWC_CentralLockArb

Type: Application SWC

Safety Level: ASIL B

Description: Arbitrates lock and unlock requests from the key fob, the automatic speed-based locking function and the crash unlock function, and produces the final central lock command with a fixed priority order.

Design Rationale: Crash unlock has the highest priority, followed by key fob requests and finally automatic locking above the speed threshold.

Runnables: Run_CentralLockArb_20ms (20 ms)

Provided Ports:

```
- P_LockCmd | SR_LockCommand | Sender-Receiver | period 20 ms
```

Required Ports:

```
- R_KeyCmd | SR_KeyFobCommand | Sender-Receiver | timeout 200 ms
- R_VehicleSpeed | SR_VehicleSpeed | Sender-Receiver | timeout 50 ms
- R_CrashStatus | SR_CrashStatus | Sender-Receiver | timeout 100 ms
```

Depends On: SWC_KeylessEntryMgr

Error Handling: On loss of the vehicle speed signal the automatic locking function is disabled and the last manual state is kept.

Verification: TC_BCM_020, TC_BCM_021

4.3 SWC_DoorLockCtrl

Component: SWC_DoorLockCtrl

Type: Application SWC

Safety Level: ASIL B

Description: Drives the door lock actuators according to the arbitrated lock command and reports the measured door lock state. It monitors actuator current to detect a blocked lock motor.

Design Rationale: Separating actuator control from arbitration keeps the safety-relevant output stage small and easy to verify.

Runnables: Run_DoorLockCtrl_10ms (10 ms)

Provided Ports:

```
- P_LockState | SR_DoorLockState | Sender-Receiver | period 50 ms
```

Required Ports:

```
- R_LockCmd | SR_LockCommand | Sender-Receiver | timeout 40 ms
- R_DoorAjar | SR_DoorAjarState | Sender-Receiver | timeout 100 ms
- R_ChildLock | SR_ChildLockState | Sender-Receiver | timeout 300 ms
```

Depends On: SWC_CentralLockArb

Error Handling: A blocked lock motor is reported after three consecutive failed attempts and the output is disabled.

Verification: TC_BCM_011, TC_BCM_012

4.4 SWC_SpeedSignalAdapter

Component: SWC_SpeedSignalAdapter

Type: Application SWC

Safety Level: QM

Description: Converts the raw wheel speed message received from the chassis network into the vehicle speed signal used by comfort functions of the body controller.

Design Rationale: A thin adapter prevents the comfort functions from depending on the signal layout of the chassis bus.

Runnables: Run_SpeedSignalAdapter_10ms (10 ms)

Provided Ports:

```
- P_VehicleSpeed | SR_VehicleSpeed | Sender-Receiver | period 10 ms
```

Required Ports:

```
- none
```

Depends On: none

Error Handling: If the chassis message times out the vehicle speed is set to zero and flagged invalid.

Verification: TC_BCM_030

4.5 SWC_BcmGateway

Component: SWC_BcmGateway

Type: Application SWC

Safety Level: QM

Description: Collects the door lock status and forwards it to the vehicle network gateway so that other ECUs can display the lock state.

Design Rationale: Keeping the gateway function inside the body controller avoids additional wiring to the cluster.

Runnables: Run_BcmGateway_100ms (100 ms)

Provided Ports:

```
- P_GwStatus | SR_BcmStatus | Sender-Receiver | period 100 ms
```

Required Ports:

```
- R_LockState | SR_DoorLockState | Sender-Receiver | timeout 150 ms
```

Depends On: SWC_DoorLockCtrl

Error Handling: When the lock state is not refreshed in time the gateway reports the state as UNKNOWN.

Verification: TC_BCM_040

4.6 SWC_ChildLockMgr

Component: SWC_ChildLockMgr

Type: Application SWC

Safety Level: ASIL B

Description: Manages the child safety lock of the rear doors and publishes whether the child lock is active.

Design Rationale: The child lock state is kept in non-volatile memory and restored after every ignition cycle.

Runnables: Run_ChildLockMgr_100ms (100 ms)

Provided Ports:

```
- P_ChildLock | SR_ChildLockState | Sender-Receiver | period 100 ms
```

Required Ports:

```
- none
```

Depends On: none

Error Handling: If the stored value is corrupted the child lock defaults to ACTIVE.

Verification: TC_BCM_050

5 Interface Definitions

All interfaces are Sender-Receiver interfaces unless stated otherwise.

5.1 SR_KeyFobCommand

Interface: SR_KeyFobCommand

Paradigm: Sender-Receiver

Data Elements: SIG_KeyFobCmd

Description: Carries the decoded key fob command.

5.2 SR_VehicleSpeed

Interface: SR_VehicleSpeed

Paradigm: Sender-Receiver

Data Elements: SIG_VehicleSpeed

Description: Carries the vehicle speed in km/h.

5.3 SR_LockCommand

Interface: SR_LockCommand

Paradigm: Sender-Receiver

Data Elements: SIG_LockCmd

Description: Carries the arbitrated central lock command.

5.4 SR_DoorLockState

Interface: SR_DoorLockState

Paradigm: Sender-Receiver

Data Elements: SIG_DoorLockState

Description: Carries the measured door lock state.

5.5 SR_DoorAjarState

Interface: SR_DoorAjarState

Paradigm: Sender-Receiver

Data Elements: SIG_DoorAjarState

Description: Carries the door ajar switch state.

5.6 SR_BcmStatus

Interface: SR_BcmStatus

Paradigm: Sender-Receiver

Data Elements: SIG_BcmStatus

Description: Carries the aggregated status sent to the gateway.

5.7 SR_ChildLockState

Interface: SR_ChildLockState

Paradigm: Sender-Receiver

Data Elements: SIG_ChildLockState

Description: Carries the child safety lock state of the rear doors.

6 Signal Catalogue

```
Signal | Data Type | Unit | Range | Update Period
SIG_KeyFobCmd | uint8 | - | 0..7 | 50 ms
SIG_VehicleSpeed | uint16 | km/h | 0..300 | 10 ms
SIG_LockCmd | uint8 | - | 0..3 | 20 ms
SIG_DoorLockState | uint8 | - | 0..3 | 50 ms
SIG_DoorAjarState | uint8 | - | 0..1 | 100 ms
SIG_BcmStatus | uint16 | - | 0..65535 | 100 ms
SIG_ChildLockState | uint8 | - | 0..1 | 100 ms
```

7 Functional Flows

The following flows describe how the components cooperate.

7.1 Remote Unlock Flow

Step 1 - SWC_KeylessEntryMgr decodes the key fob frame and publishes SIG_KeyFobCmd every 50 ms.

Step 2 - SWC_CentralLockArb receives SIG_KeyFobCmd, applies the priority rules and publishes SIG_LockCmd.

Step 3 - SWC_DoorLockCtrl drives the lock actuators and publishes SIG_DoorLockState.

Step 4 - SWC_BcmGateway forwards the lock state to the vehicle network.

7.2 Automatic Lock Above Speed Flow

Step 1 - SWC_SpeedSignalAdapter publishes SIG_VehicleSpeed every 10 ms.

Step 2 - SWC_CentralLockArb compares the speed with the locking threshold of 15 km/h.

Step 3 - When the threshold is exceeded and all doors are closed a lock command is issued.

Step 4 - SWC_DoorLockCtrl executes the command and reports the new lock state.

7.3 Crash Unlock Flow

Step 1 - A crash status request is received from the restraint system through the R_CrashStatus port.

Step 2 - SWC_CentralLockArb overrides all other requests and commands an unlock of all doors.

Step 3 - SWC_DoorLockCtrl unlocks the doors and the unlock is held until the next ignition cycle.

8 Timing and Resource Constraints

All runnables are mapped to periodic OS tasks. The 10 ms task has the highest priority. The worst-case CPU load of the door and access control functions is estimated at 12 percent of the available core capacity.

9 Verification Overview

SWC_KeylessEntryMgr is verified by TC_BCM_001, TC_BCM_002.

SWC_CentralLockArb is verified by TC_BCM_020, TC_BCM_021.

SWC_DoorLockCtrl is verified by TC_BCM_011, TC_BCM_012.

SWC_SpeedSignalAdapter is verified by TC_BCM_030.

SWC_BcmGateway is verified by TC_BCM_040.

SWC_ChildLockMgr is verified by TC_BCM_050.

10 Revision History

Revision 1.0 - Initial release of the door and access control high-level design.

Revision 1.1 - Added SWC_ChildLockMgr, interface SR_ChildLockState and signal SIG_ChildLockState. Increased the SWC_BcmGateway lock state timeout to 150 ms. Reduced the SWC_DoorLockCtrl lock state period to 50 ms. Added verification cases for SWC_CentralLockArb.